

JORGE PETIT-LAURENT SÁEZ

Energy Engineer, Industrial Engineer and MSc in Data Science. I work where the power system meets data: I model the operation of the Chilean National Electric System to evaluate storage and renewable generation projects, build cloud data infrastructure, and research climate risk and power grid resilience.

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PROFESSIONAL EXPERIENCE

Sherpas Group SpA

Energy Data Scientist

Jun 2025 – Present | Remote, Chile

Carbon markets and energy consultancy specialized in projects under Article 6 of the Paris Agreement.

- Founded the company's data area and lead its technical work in the power sector. Built from scratch the cloud infrastructure (**Google BigQuery**, **Cloud Run** and **Neo4j**) the team now uses to work with data from the Chilean National Electric Coordinator (CEN).
- Designed and coded in **Python** and **SQL** the emission-reduction methodology for battery energy storage systems (**BESS**) under **Article 6.2**. The model simulates, hour by hour and unit by unit, the **economic dispatch** of the National Electric System (SEN) on the actual **merit order**, including the **nodal marginal cost** at each busbar and charging from **renewable curtailment**.
- Turned a slow, error-prone manual Excel process into an automated, auditable and reproducible pipeline with **Git** version control and automated tests. It now supports the ex-ante and ex-post evaluation of the company's BESS portfolio, with more than 1 GW of projects.
- Defended the technical integrity of the model in **validation** processes before EBP Schweiz, Verifit, the Swiss KfK/FOEN programme and the Chilean Ministry of Environment, through data architecture reports, responses to findings and **emission factor** technical annexes.
- Developed a **FAIR** data platform for the SEN that integrates sources from CEN (SIP, Infotécnica and PLABACOM), the National Energy Commission (CNE) and the Water Directorate (DGA) into a single physical topology of unit, plant, busbar and substation. To reconcile asset names across sources I implemented identity resolution with fuzzy matching, **embeddings** and **large language models (LLM)**, and on top of it built certified indicators for curtailment, **BESS arbitrage**, **transmission congestion** and **ancillary services**.
- Deployed to production a natural-language analyst agent over the SEN, through a **Model Context Protocol (MCP)** server on Google Cloud Run, which answers with certified metrics and cites its source. The team uses it on client projects, for example to calculate the grid emission factor for a wind farm.
- Lead the **MRV** and crediting-period renewal of a 33 MW wind farm under the **Cercarbono** standard (**ACM0002** methodology and **TOOL07**): validated the CEN generation series, reconciled the plant's metering with the Coordinator's, and act as the project's technical advisor before the verifier TÜV SÜD.
- Built interactive dashboards in **Looker**, **Colab Enterprise** and **Firebase** to present results to validators, authorities and clients.

Energy Transition Center, Universidad Adolfo Ibáñez

Research Assistant, TRL Level Up Project

Oct 2023 – May 2025 | Peñalolén, Chile

Project on non-invasive inspection of photovoltaic modules, funded by AtaMoSTeC (CORFO) and SERC Chile.

- Took part in the design and execution of outdoor **daylight photoluminescence (DPL)** campaigns on photovoltaic modules connected

PUBLICATIONS

The future of atmospheric heatwaves in Chile projected by a regional climate model

Petit-Laurent, J., Martínez-Villalobos, C., Goubanova, K., Chadwick, C. and Nabat, P. (2026). *International Journal of Climatology*. doi.org/10.1002/joc.70567

Substation-level heat-wave vulnerability of the Chilean grid under global warming

Petit-Laurent, J., Martínez-Villalobos, C. and Goubanova, K. (2026). *2026 IEEE EEEIC / I&CPS Europe*, Lisbon, Portugal.

CONFERENCES AND TALKS

26th IEEE EEEIC / I&CPS Europe (2026)

Oral presentation. Lisbon, Portugal.

2nd International Symposium on Climate and Resilience, SICyR (Nov 2025)

Oral presentation: "The future of heatwaves in Chile".

5th Conference on Energy, Efficiency and Environmental Sustainability, CEES (Nov 2025)

Oral presentation: "Effects of heatwaves on electricity demand in Chile under climate change".

7th Congress of Physical Oceanography, Meteorology and Climate of the Southeast Pacific (Nov 2024)

Poster: "A projection of the future of atmospheric heatwaves in Chile".

1st Data Observatory Data Science Congress (Nov 2023)

Poster.

AWARDS AND SCHOLARSHIPS

International mobility fellowship (2026)

Research stay at the Università della Calabria (UNICAL), Italy, funded by Next Generation EU.

Data Observatory Thesis Grant (2025–2026)

Funding for the MSc research.

FEI Research Grant, UAI (2025)

Awarded for the project "The future of heatwaves in Chile under climate change".

"La Ruta del Sol" Mobility Programme, AtaMoSTeC and SERC Chile (2024)

Selected among the programme's top students.

Full Santander Open Academy scholarship (2023)

to an **inverter**, operating them at open circuit and at several power levels to reproduce **curtailment** conditions.

- Developed in **Python** and **OpenCV** a **computer vision** pipeline for the images of an InGaAs SWIR camera: cropping, perspective correction, outlier removal and normalization, to compare operating states pixel by pixel.
- Recorded the environmental variables of each campaign (irradiance, module temperature and **IV curves**) and contributed to the analysis showing that module defects can be detected even under curtailment and low irradiance, which widens the operating window for in-plant inspections.

Vivero Pilmaiquén

Plant Nursery Manager

Jan 2023 – Feb 2023 | La Unión, Chile

- Ran the nursery's daily operations during the summer season and led a team of four people.

RESEARCH

Universidad Adolfo Ibáñez and Data Observatory

MSc thesis researcher: heatwaves and electricity demand under climate change

Mar 2025 – Aug 2026 | Peñalolén, Chile

Thesis "Effects of heat waves on electricity demand in Chile in a climate change context". Funded by Data Observatory (ANID) and the 2025 UAI FEI grant.

- Developed an integrated pipeline that quantifies the sensitivity of electricity demand to heatwaves at 397 SEN substations, combining high-resolution hourly climate projections (12.5 km) with hourly energy withdrawals by busbar and customer from 2017 to 2025.
- Bias-corrected the CNRM-ALADIN64 regional climate model with **Unbiased Quantile Mapping** against ERA5-Land, and built an hourly heatwave and cold-wave climatology from 1979 to 2100 under **SSP3-7.0**, which projects heatwave frequency to grow by a factor of about 2.5 by the end of the century.
- Fitted in **R (mgcv)**, on the faculty's **HPC** cluster, 2,529 **generalized additive models (GAM)** with tensor-product interactions, one per busbar and customer. They showed that the thermal response of demand differs across climatic macro-zones and between regulated and free customers, and that the climate change impact concentrates in extreme events and in the Norte Grande region, a pattern a national aggregate analysis does not reveal.

TEACHING

Universidad Adolfo Ibáñez

Teaching Assistant

Mar 2022 – Jun 2025 | Peñalolén, Chile

- Served as teaching assistant in seven courses over three years, with more than 300 students. Led tutorial sessions, reviewed code and graded in courses such as Programming, Integral Calculus and Introduction to Engineering.

Universidad Adolfo Ibáñez

Academic Tutor

Mar 2023 – Dec 2026 (non-continuous) | Peñalolén, Chile

- Provide academic support to students through personalized study strategies and reinforcement sessions, currently in the **Energy Markets** course.

English Language Program summer school at the University of Pennsylvania, United States.

Undergraduate research award (2022)

Among the top three undergraduate research projects at UAI.

TID Scholarship (2022)

Both semesters, for undergraduate research.

COURSES AND CERTIFICATIONS

English Language Program (2023)

Institute for Academic and Professional Studies, University of Pennsylvania.

Acciona Academy Program (2024)

Renewable energy asset development and sustainability.

Microsoft and LinkedIn (Mar 2024)

Career Essentials in Data Analysis; Career Essentials in Sustainable Technology.

Python for Data Science

Pontificia Universidad Católica de Chile (Coursera).

LANGUAGES

Spanish: native

English: upper-intermediate (B2)

MEMBERSHIPS

Institute of Electrical and Electronics

Engineers (IEEE), member no. 100678726.

VOLUNTEERING

Native forest reforestation (2023 – Present)

Independent volunteer in native species planting and conservation days.

Visionarios (2021 – 2023)

Student leadership organization. Promoted and delivered leadership workshops for students in southern Chile.

Cubo Educativo (2022)

Remote programming tutor for preschool children.

EDUCATION

Universidad Adolfo Ibáñez

MSc in Data Science (Magíster en Ciencia de Datos)

Mar 2025 – Aug 2026 | Peñalolén, Chile

- Final grade: 6.51/7.0 (Chilean scale; 4.0 = pass).

Universidad Adolfo Ibáñez

Professional Degree in Energy Engineering (Ingeniero Civil en Energía), dual degree with Industrial Engineering (Ingeniero Civil Industrial)

Mar 2021 – Dec 2025 | Peñalolén, Chile

- Final grade in Energy Engineering: 6.40/7.0.
- UAI Academic Excellence Scholarship (2021–2026) and Academic Honor Scholarship (2022–2025).

Tecnológico de Monterrey

Academic exchange

2022 | Mexico